PR4 Worked Examples: Polynomial Long Division

Recording companion for the PR4 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What do I multiply the divisor by to match the leading term? Divide the leading term by the divisor's leading term to get the next quotient term. Multiply the whole divisor by it, subtract every term, and bring down the next term. Repeat until the bottom line has a lower degree than the divisor; that line is the remainder. Write the answer as quotient plus remainder over divisor, and check it by multiplying back.

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Example 1: Divide

$$x^2 + x - 6 \overline{)\quad 5x^3 - 15x^2 - 49x + 121}$$

Plotted data: A long division setup. The divisor, x squared plus x minus 6, is to the left of the division bracket. Under the bracket is the dividend, 5x cubed minus 15x squared minus 49x plus 121. The space above the bracket, where the quotient goes, is empty.

Question

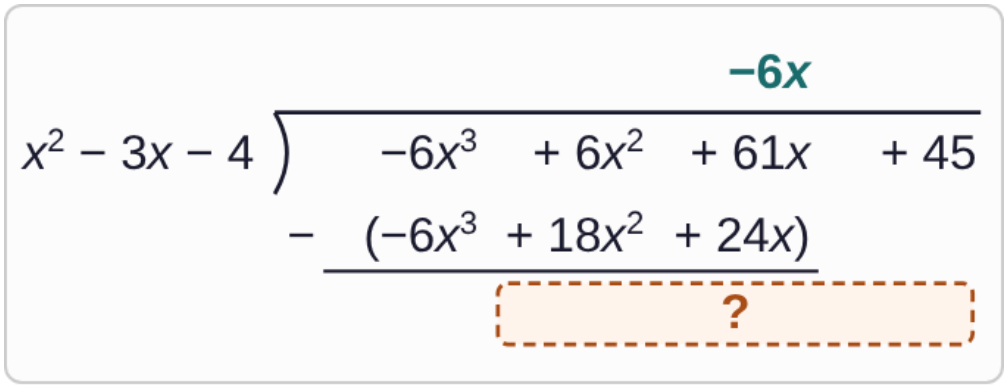
What is the first term of the quotient?

- **A.** $5x^2$
- **B.** $5x$
- **C.** 5
- **D.** $5x^3$

Answer and reasoning

Divide the leading terms: $(5x^3)/x^2=5x$, since dividing powers subtracts the exponents, $3-2=1$. Check: $5x \cdot x^2=5x^3$, the leading term. $5x^2$ divides by x instead of x^2; 5 drops the x; and $5x^3$ does not divide at all. The other terms of the dividend wait their turn. ***5x*.**

Example 2: Multiply and Subtract



Plotted data: A long division in progress. The divisor x squared minus 3x minus 4 is to the left of the bracket, and the dividend under it is negative 6x cubed plus 6x squared plus 61x plus 45. Above the bracket, the first quotient term is negative 6x. Below the dividend, marked to be subtracted, is the product negative 6x cubed plus 18x squared plus 24x, with a line under it. Below the line is an empty box with a question mark.

Question

Subtract, then bring down the 45. What goes in the box?

- **A.** $24x^2+85x+45$
- **B.** $24x^2+37x+45$
- **C.** $-12x^2+37x+45$
- **D.** $-12x^2+85x+45$

Answer and reasoning

Subtract every term: $6x^2-18x^2=-12x^2$ and $61x-24x=37x$. The $-6x^3$ terms cancel, as the leading terms always should. Then bring down 45. Adding $18x^2$ gives $24x^2$, and adding $24x$ gives $85x$; both happen when only the first term's sign gets changed. Tip: change every sign in the bottom row, then add. ***−12x²+37x+45*.**

Example 3: Divide Again

$$
\begin{array}{r}
 \quad 4x \\
x^2 - x - 6 \;\overline{)\; 4x^3 + 16x^2 - 43x - 121} \\
-\;\; (4x^3 - 4x^2 - 24x) \\
\hline
\quad 20x^2 - 19x - 121
\end{array}
$$

Plotted data: A long division in progress. The divisor x squared minus x minus 6 is to the left of the bracket, and the dividend under it is 4x cubed plus 16x squared minus 43x minus 121. Above the bracket, the first quotient term is 4x. Below the dividend, marked to be subtracted, is 4x cubed minus 4x squared minus 24x, with a line under it. Below the line is the new line, 20x squared minus 19x minus 121.

Question

What is the next term of the quotient?

- **A.** 20
- **B.** $20x$
- **C.** -20
- **D.** $20x^2$

Answer and reasoning

Divide the new leading term by the divisor's leading term: $(20x^2)/x^2=20$. It is a number, with no x, because the powers match. $20x$ divides by x instead of x^2; -20 changes the sign; and $20x^2$ skips the division. The quotient so far is $4x+20$. **20.**

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Example 4: Stop or Go On?

$$
\begin{array}{rrrrr}
 & & & 3x & -6 \\
\cline{4-5}
x^2-3x-4 \;\big)\; & 3x^3 & -15x^2 & +7x & +19 \\
- & (3x^3 & -9x^2 & -12x) & \\
\cline{2-4}
 & & -6x^2 & +19x & +19 \\
- & & (-6x^2 & +18x & +24) \\
\cline{3-5}
 & & & x & -5
\end{array}
$$

Plotted data: A long division. The divisor x squared minus 3x minus 4 is to the left of the bracket, and the dividend is 3x cubed minus 15x squared plus 7x plus 19. Above the bracket is the quotient, 3x minus 6. Below the dividend: subtract 3x cubed minus 9x squared minus 12x, leaving negative 6x squared plus 19x plus 19; then subtract negative 6x squared plus 18x plus 24, leaving x minus 5 on the bottom line.

Question

The bottom line is x − 5. Should the division stop here?

- **A.** No: keep dividing until the bottom line is just a single number.
- **B.** No: x goes into x once more, so the remainder is really −5.
- **C.** Yes: the last term of the dividend, 19, has been brought down.
- **D.** Yes: x − 5 has degree 1, less than the divisor's degree 2.

Answer and reasoning

Stop when the bottom line's degree is less than the divisor's. $x-5$ has degree 1 and x^2-3x-4 has degree 2, and x^2 does not go into x as a polynomial term. So $x-5$ is the remainder and $3x-6$ is the quotient. Bringing down the last term has to happen, but it is not the test, and a remainder can be an expression like $x-5$, not only a number. **Yes: degree 1 is less than degree 2.**

Example 5: Divide 1

$$(3x^3-6x^2-56x+62)/(x^2-6x+5)$$

Question

Use long division. Which is the result?

- **A.** $f(x)=3x-12+(x+2)/(x^2-6x+5)$
- **B.** $f(x)=3x+12+(x^2-6x+5)/(x+2)$
- **C.** $f(x)=3x+12+(x+2)/(x^2-6x+5)$
- **D.** $f(x)=3x+12+(x-2)/(x^2-6x+5)$

Answer and reasoning

$(3x^3)/x^2=3x$. Subtract $3x(x^2-6x+5)=3x^3-18x^2+15x$: $-6x^2+18x^2=12x^2$ and $-56x-15x=-71x$; bring down 62. Next, $(12x^2)/x^2=12$. Subtract $12(x^2-6x+5)=12x^2-72x+60$: $-71x+72x=x$ and $62-60=2$. The remainder $x+2$ has degree 1, so stop. The quotient $3x-12$ loses the sign in $-6-(-18)$; the remainder $x-2$ slips the sign in $62-60$; and the remainder goes over the divisor, not the other way around. ***f(x)=3x+12+(x+2)/(x²−6x+5).***

Example 6: Divide 2

$$(-3x^3-12x^2+34x+89)/(x^2+7x+10)$$

Question

Use long division. Which is the result?

- **A.** $f(x)=-3x+9+(x-1)/(x^2+7x+10)$
- **B.** $f(x)=3x+9+(x-1)/(x^2+7x+10)$
- **C.** $f(x)=-3x-9+(x-1)/(x^2+7x+10)$
- **D.** $f(x)=-3x+9+x-1$

Answer and reasoning

$(-3x^3)/x^2=-3x$. Subtract $-3x(x^2+7x+10)=-3x^3-21x^2-30x$: $-12x^2+21x^2=9x^2$ and $34x+30x=64x$; bring down 89. Next, $(9x^2)/x^2=9$. Subtract $9x^2+63x+90$: $64x-63x=x$ and $89-90=-1$. The remainder is $x-1$. The quotient's first term keeps the negative sign; -9 comes from $-12-21$ instead of $-12-(-21)$; and the remainder has to be divided by the divisor, not just added on. ***f(x)=−3x+9+(x−1)/(x²+7x+10).***

Example 7: Divide 3

$$x^2 + 3x - 4 \;\overline{)\; 2x^3 - 25x + 26 \;}\qquad \text{quotient: } 2x$$

Plotted data: A long division setup. The divisor x squared plus 3x minus 4 is to the left of the division bracket. Under the bracket is the dividend, written 2x cubed minus 25x plus 26. Above the bracket, the first term of the quotient, 2x, has been written.

Question

The dividend has no x^2 term. Multiply 2x by the divisor and subtract. What is the new line, after bringing down?

- **A.** $-31x^2+34x$
- **B.** $-6x^2-33x+26$
- **C.** $6x^2-17x+26$
- **D.** $-6x^2-17x+26$

Answer and reasoning

Rewrite the dividend with a 0 for the missing power, $2x^3+0x^2-25x+26$, so like terms line up. Subtract $2x(x^2+3x-4)=2x^3+6x^2-8x$: $0x^2-6x^2=-6x^2$ and $-25x-(-8x)=-17x$; bring down 26. Without the 0, the $6x^2$ lands under $-25x$, which gives $-31x^2+34x$. The $6x^2$ version adds instead of subtracting, and $-33x$ adds $-8x$ instead of subtracting it. Finishing gives $f(x)=2x-6+(x+2)/(x^2+3x-4)$. $\mathbf{-6x^2-17x+26}$**.**

Example 8: Divide 4

$$(-3x^3+3x^2+67x-119)/(x^2+3x-10)$$

Question

Use long division. What is the quotient?

- **A.** $-3x-12$
- **B.** $-3x+12$
- **C.** $-3x^2+12x$
- **D.** $3x+12$

Answer and reasoning

$(-3x^3)/x^2=-3x$. Subtract $-3x(x^2+3x-10)=-3x^3-9x^2+30x$: $3x^2+9x^2=12x^2$ and $67x-30x=37x$; bring down -119. Next, $(12x^2)/x^2=12$, and subtracting $12x^2+36x-120$ leaves $x+1$, degree 1, the remainder. -12 comes from $3-9$ instead of $3-(-9)$; $3x+12$ drops the negative; and $-3x^2+12x$ divides by x instead of x^2. A degree 3 polynomial divided by a degree 2 one has a degree 1 quotient. **$-3x+12$, with remainder $x+1$.**

Example 9: Write It

$$(-2x^3+6x^2+21x-47)/(x^2-6x+8)$$

Question

Long division gives quotient −2x − 6 and remainder x + 1. How does CheckIt want the answer written?

A. $f(x)=-2x-6+x+1$
B. $f(x)=-2x-6+(x+1)/(x^2-6x+8)$
C. $f(x)=-2x-6+(x^2-6x+8)/(x+1)$
D. $f(x)=(-2x-6)/(x^2-6x+8)+x+1$

Answer and reasoning

It works like $17\div 5=3+2/5$: the quotient, plus the remainder over the divisor. Adding $x+1$ on its own leaves out the division; flipping the fraction puts the divisor on top; and the quotient is never divided by anything. ***f(x)=−2x−6+(x+1)/(x²−6x+8).***

Example 10: Check It

$$f(x)=(-4x+8)(x^2-4x-5)+(x+3)$$

Question

Dividing f(x) by x² − 4x − 5 gave quotient −4x + 8 and remainder x + 3. Multiply back as shown. What is f(x)?

A. $-4x^3+24x^2-12x-40$

B. $-4x^3+24x^2-13x-43$

C. $-4x^3+24x^2-11x-37$

D. $-4x^3+16x^2-11x-37$

Answer and reasoning

$(-4x+8)(x^2-4x-5)=-4x^3+16x^2+20x+8x^2-32x-40=-4x^3+24x^2-12x-40$. Adding $x+3$ gives $-4x^3+24x^2-11x-37$. Stopping before the remainder is added, subtracting it, or missing the $8x^2$ from $8\cdot x^2$ gives the other three. Use this check on your own CheckIt answer: if it does not give back the dividend, look for a sign slip. **$-4x^3+24x^2-11x-37$.**

Example 11: Predict

$$(ax^3+bx^2+cx+d)/(x^2+px+q)$$

Question

Before you start: dividing a degree 3 polynomial by a degree 2 polynomial, what degrees can the quotient and the remainder have?

A. Quotient of degree 1; remainder of degree 1 or less
B. Quotient of degree 1; remainder of degree 2 or less
C. Quotient of degree 2; remainder of degree 1 or less
D. Quotient of degree 0; remainder of degree 1 or less

Answer and reasoning

The first quotient term is $x^3 \div x^2 = x$, so the quotient has degree $3-2=1$. The division stops as soon as the bottom line's degree drops below 2, so the remainder has degree 1 or less. A degree 2 remainder would mean you could still divide again. That is the shape of every CheckIt answer: a degree 1 quotient, plus a degree 1 remainder over the divisor. **Quotient of degree 1; remainder of degree 1 or less.**

Example 12: Look Ahead

$$f(x)=2x-2+(x-2)/(x^2+7x+12)$$

Question

This is what long division gives for one function f. When x is very large, the graph of f is very close to which line?

A. $y=2$
B. $y=0$
C. $y=3x-4$
D. $y=2x-2$

Answer and reasoning

The fraction's top has degree 1 and its bottom has degree 2, so for large x the bottom grows much faster and the fraction shrinks toward 0. At $x=100$ it is $98/10712\approx 0.009$. What is left is $2x-2$, so the graph hugs the line $y=2x-2$. In calculus this line is called a slant asymptote, and you will find it with limits. $y=2$ compares leading coefficients, which works only when the top and bottom degrees match; $y=0$ is where the fraction goes, not the whole function; and $3x-4$ adds the remainder on whole instead of keeping it over the divisor. ***y=2x−2*.**

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Example 13: Slip 1

$$
\begin{array}{r|l}
 & \quad 4x \\
\cline{2-2}
x^2 - 2x - 3 & \quad 4x^3 + 16x^2 - 59x - 74 \\
 & -\ (4x^3 - 8x^2 - 12x) \\
 & \quad 24x^2 - 71x - 74
\end{array}
$$

Plotted data: A student's long division. The divisor x squared minus 2x minus 3 is to the left of the bracket, and the dividend is 4x cubed plus 16x squared minus 59x minus 74. Above the bracket, the first quotient term is 4x. Below the dividend, marked to be subtracted, is 4x cubed minus 8x squared minus 12x, with a line under it. The student's new line below is 24x squared minus 71x minus 74.

Question

A student is dividing $(4x^3 + 16x^2 - 59x - 74)$ by $(x^2 - 2x - 3)$. Here is the first subtraction. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** $16x^2 - (-8x^2)$ is $8x^2$, so the line should start with $8x^2$.
- **C.** $-12x$ was added: $-59x - (-12x)$ is $-47x$, not $-71x$.
- **D.** The first term of the quotient should be $4x^2$, not $4x$.

Answer and reasoning

Subtracting changes every sign in the bottom row. $16x^2-(-8x^2)=24x^2$, which the student got right, but $-59x-(-12x)=-47x$, not $-71x$. The new line is $24x^2-47x-74$, and $4x^3 \div x^2=4x$ was right. **The *x*-term was added instead of subtracted.**

Example 14: Slip 2

$$f(x)=4x+(8x^2-39x-51)/(x^2-5x-6)$$

Question

For $(4x^3 - 12x^2 - 63x - 51) \div (x^2 - 5x - 6)$, a student gave quotient $4x$ and remainder $8x^2 - 39x - 51$, and wrote this answer. What is the mistake, if any?

A. There is no mistake.

B. $8x^2 - 39x - 51$ has degree 2, not less than 2: divide again.

C. The quotient should be $4x^2$, since $4x^3 \div x$ is $4x^2$.

D. The remainder should be written over $4x^3 - 12x^2 - 63x - 51$.

Answer and reasoning

A remainder must have a lower degree than the divisor. $8x^2-39x-51$ has degree 2, the same as x^2-5x-6, so there is another step: $8x^2 \div x^2=8$, and subtracting $8x^2-40x-48$ leaves $x-3$. The first step, $4x^3 \div x^2=4x$, was right. **Quotient $4x+8$, remainder $x-3$.**

Example 15: Slip 3

$$(5x+10)(x^2-4x+3)$$
$$+(x+4)$$
$$=5x^3-10x^2-25x+30$$
$$+x+4$$
$$=5x^3-10x^2-24x+34$$

Question

For $(5x^3 - 10x^2 - 24x + 34) \div (x^2 - 4x + 3)$, a student got quotient $5x + 10$ and remainder $x + 4$, then checked it this way. What is the mistake, if any?

A. There is no mistake.
B. The check should add the divisor, not the remainder.
C. The remainder should be $x - 4$, from the last subtraction.
D. The quotient should be $5x - 10$, from $-10x^2 - (-20x^2)$.

Answer and reasoning

$(5x+10)(x^2-4x+3)=5x^3-10x^2-25x+30$, and adding $x+4$ gives $5x^3-10x^2-24x+34$, the dividend. Also $-10x^2-(-20x^2)=10x^2$, so the quotient's second term is $+10$. Multiplying back like this is how to be sure before you submit. **There is no mistake.**

Example 16: Slip 4

$$f(x)=-4x+4+x-3$$

Question

For $(-4x^3 + 20x^2 + 33x - 51) \div (x^2 - 4x - 12)$, a student found quotient $-4x + 4$ and remainder $x - 3$, and wrote this answer. What is the mistake, if any?

A. There is no mistake.

B. The quotient should be $4x + 4$, since $-4x^3 \div x^2$ is a positive $4x$.

C. It is not simplified: combine like terms to get $-3x + 1$.

D. The remainder goes over the divisor: $+ (x - 3)/(x^2 - 4x - 12)$.

Answer and reasoning

The remainder is divided by the divisor, the way $17 \div 5 = 3 + 2/5$. Writing $+x-3$ adds it on whole, and combining that into $-3x+1$ only simplifies a wrong answer. $-4x^3 \div x^2 = -4x$ keeps its sign. ***$f(x)=-4x+4+(x-3)/(x^2-4x-12)$*.**

Example 17: Slip 5

$$
\begin{array}{r|l}
 & \quad 6x \\
\cline{2-2}
x^2 + x - 6 & \quad 6x^3 - 41x + 32 \\
 & -\ (6x^3 + 6x^2 - 36x) \\
 & \quad -47x^2 + 68x
\end{array}
$$

Plotted data: A student's long division. The divisor x squared plus x minus 6 is to the left of the bracket, and the dividend is written 6x cubed minus 41x plus 32. Above the bracket, the student wrote 6x over the second column. Below the dividend, marked to be subtracted, is 6x cubed plus 6x squared minus 36x, written so that 6x squared is under negative 41x and negative 36x is under 32. The student's new line is negative 47x squared plus 68x.

Question

A student is dividing $(6x^3 - 41x + 32)$ by $(x^2 + x - 6)$. Here is the first subtraction. What is the mistake, if any?

A. There is no mistake.
B. The first term of the quotient should be $6x^2$, since $6x^3 \div x$ is $6x^2$.
C. Without $0x^2$, the columns slip; the line should be $-6x^2 - 5x + 32$.
D. The product should be $6x^3 - 6x^2 + 36x$, since it is being subtracted.

Answer and reasoning

The dividend has no x^2 term, so write it as $6x^3+0x^2-41x+32$. Then $6x^2$ sits under $0x^2$ and $-36x$ under $-41x$: $0-6x^2=-6x^2$ and $-41x-(-36x)=-5x$, then bring down 32. Combining $-41x$ with $6x^2$ mixes unlike terms. The product $6x(x^2+x-6)=6x^3+6x^2-36x$ was right; the signs change when you subtract, not before. **The new line is $-6x^2-5x+32$.**